

CS 486/686

Constraint Satisfaction Problems

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Lecture 4

RN 6.1–6.3 · PM 4.1–4.4

Reminders

DUE

Tue May 26

Chat 1 (Intro & uninformed search) — due 11:59 PM.

RELEASED TODAY

Due Thu May 28

Chat 2 (Heuristic search & CSPs) — two-day extension from the original Tue May 26.

DISTINGUISHED LECTURE

Tue May 26, 10:00–11:30 AM · DC 1302

Prof. [Kyunghyun Cho](#) (NYU) — co-inventor of *attention* (Bahdanau, Cho, Bengio, ICLR 2015) and the *GRU*. **Highly recommended.**

Search ›

Uncertainty ›

Decisions ›

Learning

Learning goals

- Formulate a real-world problem as a CSP
- Trace backtracking search
- Verify whether a constraint is arc-consistent
- Trace the AC-3 algorithm
- Trace backtracking *combined* with arc consistency

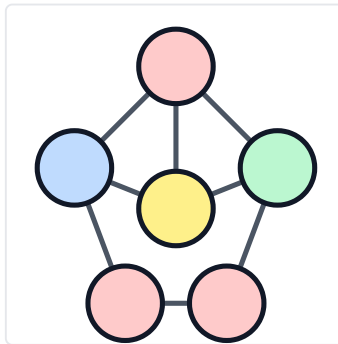
CSPs you've seen

C	A	T		S
A	R	E		
D		O	N	E

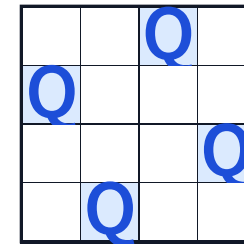
Crosswords

5	3		7			
6			1	9	5	
	9	8				6
8			6		3	
4		8		3		1
7			2		6	
	6				2	8
			4	1	9	5
			8		7	9

Sudoku



Graph colouring



N-queens

Naive idea: generate-and-test

Enumerate every assignment, test if it satisfies the constraints.

```
A=1, B=1, C=1, D=1, E=1    # fail
```

```
A=1, B=1, C=1, D=1, E=2    # fail
```

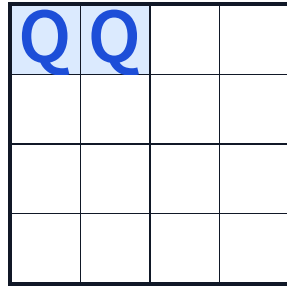
```
...
```

5 variables, domain size 4: $4^5 = 1024$ assignments. Doesn't scale.

And it's wasteful: many constraints can be checked on *partial* assignments.

Search algorithms are structure-blind

A search algorithm sees only "generate successors" and "is this a goal?".



Two queens already in the same row — clearly a dead end — yet the search keeps generating successors. We need to **see the structure**.

A CSP is a triple (X, D, C)

Variables X

$\{X_1, X_2, \dots, X_n\}$

Domains D

Each X_i draws values
from a finite set D_i .

Constraints C

Each $c \in C$ restricts
allowed combinations
of variable values.

Solution: an assignment to all variables that satisfies every constraint.

4-queens: variables and domains

- One queen per column $\Rightarrow$ track its *row*.
- **Variables:** x_0, x_1, x_2, x_3 (column index).
- **Domains:** $D_{x_i} = \{0, 1, 2, 3\}$.

	x_0	x_1	x_2	x_3
0				
1				
2				
3				

4-queens: constraints

No pair of queens in the same row or diagonal.

$$\forall i \neq j : (x_i \neq x_j) \wedge (|x_i - x_j| \neq |i - j|)$$

$$\text{Columns } 0, 1: (x_0 \neq x_1) \wedge (|x_0 - x_1| \neq 1).$$

No *column* constraint needed — our variables already encode “one queen per column”.

Q: which constraints do we need?

Q. Given the variables $x_0, \dots, x_3$ and their domains, which row/column/diagonal constraints must we enforce?

- A. No two queens in the same row
- B. No two queens in the same column
- C. No two queens on the same diagonal
- D. Two of (A), (B), (C)
- E. All of (A), (B), (C)

D — row + diagonal only. The column constraint is already implied by our choice of variables: there's exactly one x_i per column, so no two queens can ever share a column.

Two ways to express a constraint

Constraint: queens in columns 0 and 2 are not in the same row or diagonal.

As a formula

$$(x_0 \neq x_2) \wedge (|x_0 - x_2| \neq 2)$$

As a table

x_0	x_2
0	1
0	3
1	0
1	2
2	1
2	3
3	0
3	2

Q: encode (x_0, x_2) as a formula

Q. Which formula encodes "queens in columns 0 and 2 are not in the same row or diagonal"?

- A. $(x_0 \neq x_2)$
- B. $(x_0 \neq x_2) \wedge ((x_0 - x_2) \neq 1)$
- C. $(x_0 \neq x_2) \wedge ((x_0 - x_2) \neq 2)$
- D. $(x_0 \neq x_2) \wedge (|x_0 - x_2| \neq 1)$
- E. $(x_0 \neq x_2) \wedge (|x_0 - x_2| \neq 2)$

E. The two columns are 2 apart, so diagonal conflict means $|x_0 - x_2| = 2$. We need both the row and diagonal exclusions, and the absolute value because the queens could be above or below each other.

4-queens as an incremental search problem

- **State:** one queen per column in the leftmost k columns, no pair attacking each other.
- **Initial state:** empty board _ _ _ _.
- **Goal state:** 4 queens placed, no pair attacking. e.g. 1 3 0 2.
- **Successor:** place a queen in the leftmost empty column that no current queen attacks.

e.g. 0 _ _ _ $\rightarrow$ 0 2 _ _ , 0 3 _ _ .

Any search algorithm works on this state space. **Backtracking** = **DFS** + one rule: skip any partial assignment that already violates a constraint.

Backtracking search

Backtracking = DFS through partial assignments + prune any partial that already violates a constraint.

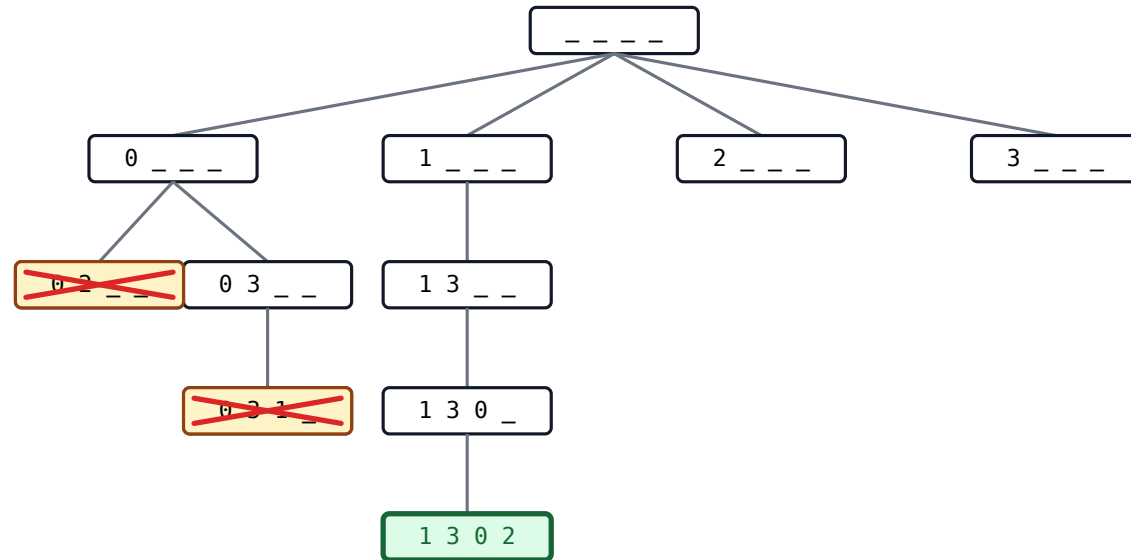
backtrack(assignment, csp)

1. If **assignment** is complete → **return** it.
2. Pick an unassigned variable **var**.
3. For each **value** in **domain(var)** that satisfies every constraint:
4. Add **{var = value}** and go deeper; if a solution is found, return it.
5. Otherwise undo and try the next value.
6. If no value worked → **return failure**.

Trace: backtracking on 4-queens

Each node = partial assignment. Yellow = dead end. Green = solution.

Step:



Found in 9 expansions.

Partial assignment, fatal future

After placing $x_0 = 0$, is $x_1 = 2$ safe?

0	Q	X	X	X
1	X	X	X	
2	X	Q	X	X
3	X	X	X	X

Every cell of column x_3 is attacked $\rightarrow$ **no future** for the remaining queens.

Knowing this *before* trying $x_2 =$ **arc consistency**.

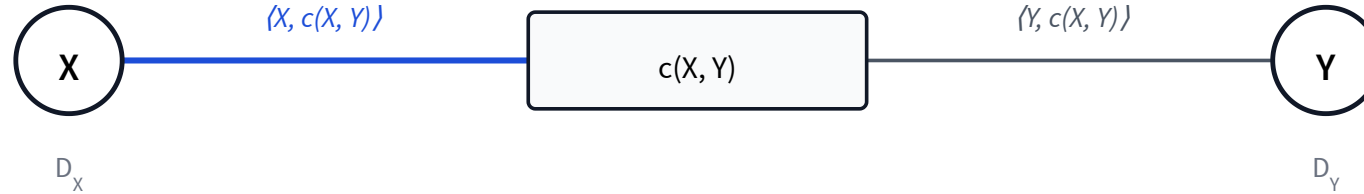
Binary constraints only

Arc consistency is defined on *binary* constraints $c(X, Y)$.

Unary constraints $c(X)$: drop disallowed values directly from D_X .

The CSPs in this course (4-queens, the homework problems) are all binary.

Notation: arcs



- An **arc** $\langle X, c(X, Y) \rangle$ pairs a constraint with one of its variables.
- X is the **primary** variable (the one we'll prune); Y is the **secondary**.
- Each binary constraint defines *two* arcs — one per primary choice.

Arc-consistency definition

Arc consistency. The arc $\langle X, c(X, Y) \rangle$ is *arc-consistent* iff for every $v \in D_X$ there exists a $w \in D_Y$ such that (v, w) satisfies $c(X, Y)$.

Intuition: every value in the primary domain has a partner in the secondary domain.

If some $v \in D_X$ has no partner, remove v from D_X to make the arc consistent.

Q: is this arc consistent?

Q. Constraint $X < Y$, with $D_X = D_Y = \{1, 2\}$. Is $\langle X, X < Y \rangle$ arc-consistent?

A. Yes

B. No

B — No. For $v = 2 \in D_X$, there is no $w \in D_Y$ with $2 < w$. We can **remove 2 from D_X** to make the arc consistent.

The AC-3 algorithm

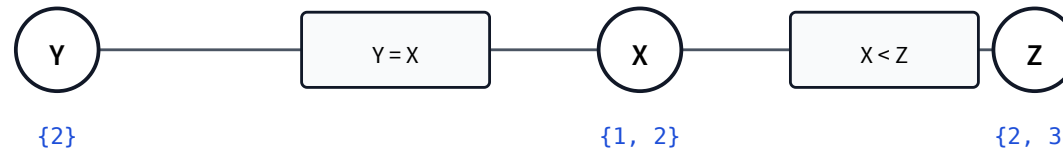
Process arcs one at a time. After each shrink, re-queue neighbours whose consistency may have broken.

AC-3(arcs)

1. $S \leftarrow$ set of all arcs. *initialise the queue*
2. While S is not empty:
 1. Pop an arc $\langle X, c(X, Y) \rangle$ from S .
 2. For each $v \in D_X$ with no $w \in D_Y$ satisfying $c(X, Y) \rightarrow$ remove v from D_X .
 3. If D_X is now empty $\rightarrow$ return false. *infeasible — no value works*
 4. If D_X shrank $\rightarrow$ for every neighbour $Z \neq Y$, add $\langle Z, c'(Z, X) \rangle$ back into S . *re-queue — its consistency may have broken*
3. Return true. *every arc is consistent*

Why re-queue arcs?

Shrinking one domain can break a previously-consistent neighbouring arc.

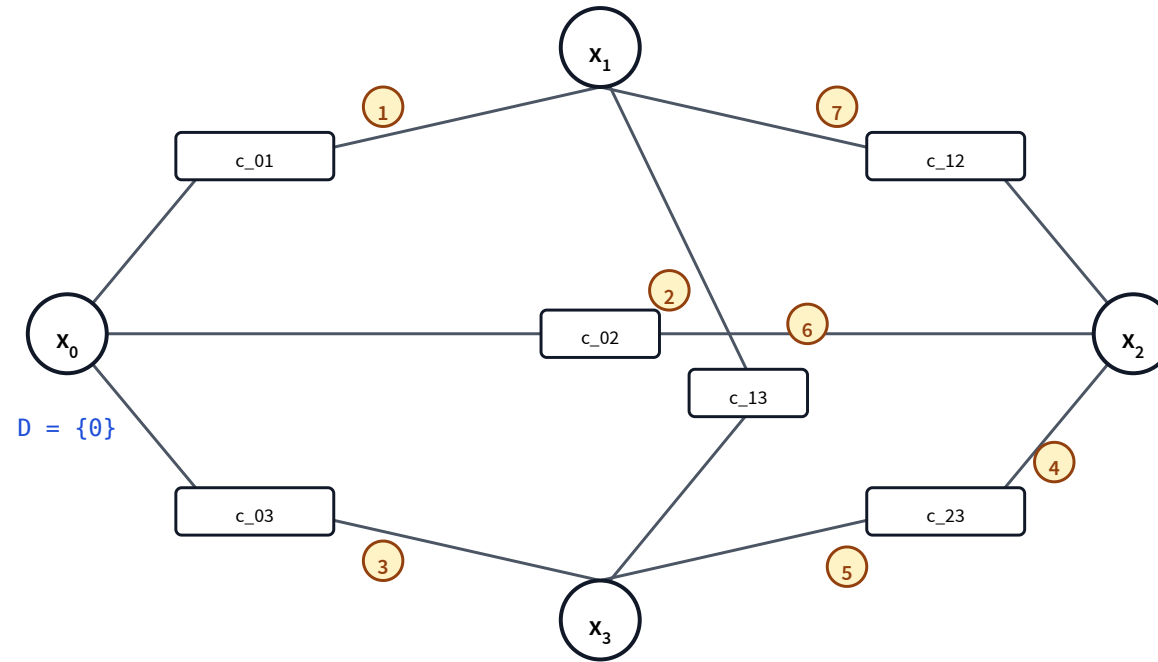


Initially: arc $\langle X, X < Z \rangle$ is consistent ($1 < 2$, $1 < 3$, $2 < 3$ all work).

Then: arc $\langle X, Y = X \rangle$ forces $D_X \rightarrow \{2\}$.

Now: $\langle Z, X < Z \rangle$ is broken — for $w = 2 \in D_Z$, no $v < 2$ remains in D_X . **Must re-queue.**

Trace: AC-3 on 4-queens with $x_0 = 0$



Three outcomes of AC-3

- A domain becomes empty → no solution.
- Every domain has exactly one value → solution found, no search needed.
- Some domains have multiple values → need backtracking search on what remains.

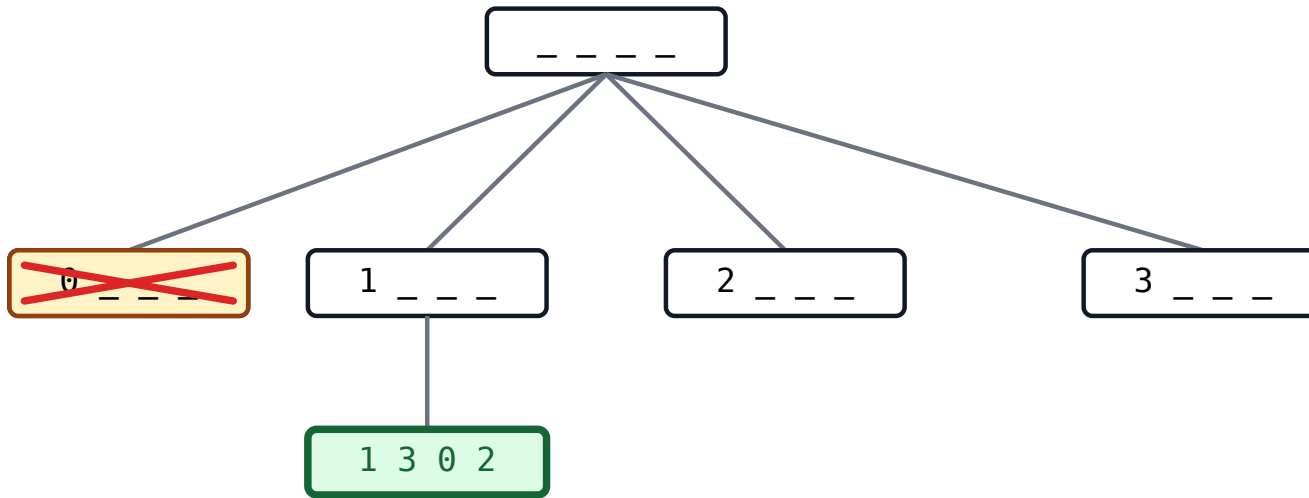
AC-3 — properties

Terminates?	Yes <i>each arc enqueued at most d times</i>
Order of arcs?	Doesn't matter <i>same final domains</i>
Time complexity	$O(c d^3)$ <i>c constraints $\times d$ re-queues per arc $\times O(d^2)$ per check</i>

Combining backtracking with AC-3

1. Run backtracking search.
2. After each value assignment, run AC-3.
3. If a domain becomes empty → **fail this branch**, backtrack.
4. If every domain has exactly one value → **solution found**.
5. Otherwise → continue backtracking on the remaining unassigned variables.

Trace: backtracking + AC-3 on 4-queens



Why combine BT with AC-3?

- **AC-3 prunes branches BT would otherwise explore** — it kills bad partial assignments before they grow.
- **BT decides what AC-3 can't** — when AC-3 leaves multi-value domains, BT picks a value and triggers another AC-3 pass.
- On 4-queens with $x_0 = 1$, AC-3 alone reduced everything to a single solution — **zero further search**.

Three techniques for CSPs

Technique	How	Finds a solution?	Best at
Backtracking	DFS through assignments, prune on constraint violation	Yes	generic, easy to implement
AC-3 alone	Repeatedly enforce arc consistency	<i>only if all domains become singletons</i>	shrinking domains preemptively
BT + AC-3	BT picks a value, AC-3 propagates	Yes <i>far fewer expansions</i>	practical CSP solving

Learning goals (recap)

- ✓ Formulate a real-world problem as a CSP
- ✓ Trace backtracking search
- ✓ Verify arc consistency
- ✓ Trace AC-3
- ✓ Combine backtracking with arc consistency